Q1 — Portfolio Overview

Why This Question?

Before any risk analysis can begin, we need to establish what we are working with. Q1 defines the scale and shape of the portfolio — total loans, total capital requested, total capital deployed, and average loan size. The comparison between requested and funded amounts also reveals the lender's disposition: are they lending freely or applying caution? This is the foundation every subsequent finding sits on. Without it, percentages and rates have no context.

```
SELECT
   COUNT(*) AS total_loans,
   SUM(loan_amnt) AS total_requested_amnt,
   SUM(funded_amnt) AS total_funded_amnt,
   ROUND(AVG(funded_amnt)::NUMERIC, 2) AS avg_funded_amnt,
   ROUND(AVG(loan_amnt)::NUMERIC, 2) AS avg_loan_amnt
FROM loans;
```

Result

total_loans	total_requested	total_funded	avg_funded	avg_loan
39,715	445,596,650	434,804,325	10,948.11	11,219.86

Interpretation The portfolio contains 39,715 loans with a total requested amount of $445.6M, of which $434.8M was actually disbursed. The average funded amount of $10,948 is $272 lower than the average requested amount of $11,220, indicating the lender consistently applied caution across the portfolio by under-funding requests a conservative lending policy designed to limit individual and aggregate exposure.

Q2 — Loan Distribution by Grade

```
SELECT
   grade,
   COUNT(*) AS loan_count,
   ROUND(COUNT(*) * 100.0::NUMERIC / (SELECT COUNT(*) FROM loans), 2) AS portfolio_share
FROM loans
GROUP BY grade
ORDER BY grade;
```

Result

Grade	Loan Count	Portfolio Share %
A	10,084	25.39
B	12,019	30.26
C	8,098	20.39
D	5,307	13.36
E	2,842	7.16
F	1,049	2.64
G	316	0.80

Interpretation Grades A and B account for 55.65% of the portfolio, indicating a predominantly low-risk borrower base and a stable lending book. However, Grades E, F, and G though only 10.6% by count cannot be dismissed, as their higher default rates and loan amounts may contribute disproportionately to portfolio losses, which Section 4 will quantify.

Why This Question Matters Grade distribution reveals the risk composition of the entire portfolio at a glance. A concentration in lower grades signals loose underwriting; a concentration in higher grades signals conservative lending. Neither extreme is without risk this analysis sets the baseline for every charge-off and exposure question that follows.

Q3 — Loan Distribution by Purpose

```
SELECT
   purpose,
   COUNT(*) AS loan_count,
   ROUND(COUNT(*) * 100.0::NUMERIC / (SELECT COUNT(*) FROM loans), 2) AS portfolio_purpose_share,
   ROUND(AVG(loan_amnt)::NUMERIC - AVG(funded_amnt)::NUMERIC, 2) AS avg_funding_gap
FROM loans
GROUP BY purpose
ORDER BY loan_count DESC;
```

Result

Purpose	Loan Count	Portfolio Share %	Avg Funding Gap
debt_consolidation	18,641	46.94	306.29
credit_card	5,130	12.92	271.46
other	3,992	10.05	174.59
home_improvement	2,976	7.49	316.76
major_purchase	2,187	5.51	185.19
small_business	1,827	4.60	551.78
car	1,549	3.90	63.22
wedding	947	2.38	123.07
medical	693	1.74	160.28
moving	583	1.47	143.22
vacation	381	0.96	83.86
house	381	0.96	358.40
educational	325	0.82	118.38
renewable_energy	103	0.26	208.25

Why This Question Matters

Loan purpose reveals what borrowers are actually doing with the money, which carries different repayment risk profiles. Debt consolidation, for example, restructures existing obligations rather than creating new productive value, while small business loans fund ventures with inherently higher failure rates.

Interpretation
Debt consolidation dominates the portfolio with 18,641 loans, representing nearly half (46.94%) of all lending activity, indicating that the majority of borrowers are using this product to restructure existing debt rather than fund new spending. Small business loans, while only 4.60% of the portfolio by count, show the largest funding gap at £551.78 below the requested amount — signalling that the lender treats this purpose with the highest degree of caution. Combined with its smaller volume, small business warrants close attention in the upcoming charge-off analysis to determine whether this caution is justified by actual default behaviour.

Q4 — Loan Term Distribution

```
SELECT
    term,
    COUNT(*) AS loan_count,
    ROUND(COUNT(*) * 100.0::NUMERIC / (SELECT COUNT(*) FROM loans), 2) AS portfolio_share
FROM loans
GROUP BY term;
```

Result

Term	Loan Count	Portfolio Share %
36 months	29,094	73.26
60 months	10,621	26.74

Why This Question Matters

Loan term affects risk exposure duration. Longer terms give more time for adverse life events — job loss, medical emergencies, income disruption — to interfere with repayment, and are typically associated with larger loan amounts, increasing monthly repayment burden.

Interpretation
The portfolio is dominated by 36-month loans, which account for 73.26% of all loans, while 60-month loans represent the remaining 26.74%. Despite their smaller share, 60-month loans typically carry larger loan amounts and longer repayment exposure, meaning their contribution to total portfolio risk could be disproportionately higher than their count suggests — this will be tested directly in Q13's charge-off rate by term.

Q5 — Overall Portfolio Charge-Off Rate

```
SELECT
    loan_status,
    COUNT(*) AS chargeoff_count,
    ROUND(COUNT(*) * 100.0::NUMERIC /
```

```
    (SELECT COUNT(*) FROM loans WHERE loan_status IN ('Charged Off', 'Fully Paid')), 2) AS chargeoff_rate
FROM loans
WHERE loan_status = 'Charged Off'
GROUP BY loan_status;
```

Result

Loan Status	Count	Charge-Off Rate %
Charged Off	5,626	14.58

Why This Question Matters

This establishes the baseline loss rate for the entire portfolio — the single most important risk metric. Every subsequent question in this section breaks this number down by borrower segment to find where the risk concentrates.

Interpretation

The portfolio's overall charge-off rate of 14.58% — roughly one in every seven loans — is significantly above the 3-5% range typically considered healthy for consumer lending, indicating material credit risk within this book. This elevated rate justifies the deeper segment-level analysis in the following sections to identify which borrower characteristics are driving this loss rate.

Q6 — Charge-Off Rate by Grade

```
SELECT
   grade,
   chargeoff_count,
   resolved_chargeoff,
   ROUND(chargeoff_count * 100.0::NUMERIC / resolved_chargeoff, 2) AS chargeoff_rate
FROM (
   SELECT
      grade,
      COUNT(CASE WHEN loan_status = 'Charged Off' THEN 1 END) AS chargeoff_count,
      COUNT(CASE WHEN loan_status IN ('Charged Off', 'Fully Paid') THEN 1 END) AS resolved_chargeoff
   FROM loans
   WHERE loan_status IN ('Charged Off', 'Fully Paid')
   GROUP BY grade
) AS chargeoff
ORDER BY grade;
```

Result

Grade	Charge-Off Count	Resolved Count	Charge-Off Rate %
A	602	10,044	5.99
B	1,424	11,674	12.20
C	1,347	7,834	17.19
D	1,118	5,085	21.99
E	715	2,663	26.85

F	319	976	32.68
G	101	299	33.78

Why This Question Matters

This breaks the portfolio-wide 14.58% charge-off rate down by grade to test whether the lender's risk classification system actually predicts default behaviour — and to identify which grades are driving the overall loss rate.

Interpretation

Charge-off rates increase monotonically from Grade A (5.99%) to Grade G (33.78%), a spread of nearly 28 percentage points, confirming that the lender's grading system is functioning as designed — risk classification accurately predicts default likelihood. While Grades E, F, and G collectively represent only 10.6% of the portfolio by loan count, their charge-off rates run at roughly 2x to 2.3x the portfolio average of 14.58%, indicating these segments likely contribute a disproportionately large share of total dollar losses relative to their size. This concentration of risk in a small subset of the portfolio will be quantified precisely in Section 4's exposure analysis.

Q7 — Charge-Off Rate by Purpose

```
SELECT
   purpose,
   chargeoff_count,
   resolved_count,
   ROUND(chargeoff_count * 100.0::NUMERIC / resolved_count, 2) AS chargeoff_rate
FROM (
   SELECT
      purpose,
      COUNT(CASE WHEN loan_status = 'Charged Off' THEN 1 END) AS chargeoff_count,
      COUNT(CASE WHEN loan_status IN ('Fully Paid', 'Charged Off') THEN 1 END) AS resolved_count
   FROM loans
   WHERE loan_status IN ('Fully Paid', 'Charged Off')
   GROUP BY purpose
) AS rate
ORDER BY purpose;
```

Result

Purpose	Charge-Off Count	Resolved Count	Charge-Off Rate %
small_business	474	1,753	27.04
renewable_energy	19	102	18.63
educational	56	325	17.23
other	633	3,864	16.38
house	59	367	16.08
moving	92	576	15.97
medical	106	681	15.57

Purpose	Charge-Off Count	Resolved Count	Charge-Off Rate %
debt_consolidation	2,767	18,055	15.33
vacation	53	375	14.13
home_improvement	347	2,875	12.07
car	160	1,499	10.67
credit_card	542	5,027	10.78
wedding	96	926	10.37
major_purchase	222	2,150	10.33

Why This Question Matters

This breaks the charge-off rate down by loan purpose to identify which use-cases carry the highest default risk, and to verify whether the funding caution observed in Q3 was actually justified by real default behaviour.

Interpretation

Small business loans carry the highest charge-off rate in the portfolio at 27.04% — nearly double the portfolio average of 14.58% — confirming that the lender's caution observed in Q3 (the largest funding gap of £551.78) was a justified risk response rather than excessive conservatism. At the other end, major purchase loans show the lowest charge-off rate at 10.33%, indicating this purpose carries comparatively low default risk despite receiving less funding scrutiny. These findings suggest the lender should consider tightening underwriting criteria or pricing further for small business loans, while major purchase loans may represent an opportunity for more competitive terms given their strong repayment performance.

Q8 — Charge-Off Rate by Verification Status

```
SELECT
   verification_status,
   chargedoff_count,
   resolved_count,
   ROUND(chargedoff_count * 100.0::NUMERIC / resolved_count, 2) AS chargeoff_rate
FROM (
   SELECT
      verification_status,
      COUNT(CASE WHEN loan_status = 'Charged Off' THEN 1 END) AS chargedoff_count,
      COUNT(CASE WHEN loan_status IN ('Fully Paid', 'Charged Off') THEN 1 END) AS resolved_count
   FROM loans
   WHERE loan_status IN ('Fully Paid', 'Charged Off')
   GROUP BY verification_status
) AS chargeoff_rate
ORDER BY chargeoff_rate DESC;
```

Result

Verification Status	Charge-Off Count	Resolved Count	Charge-Off Rate %
Verified	2,051	12,206	16.80
Source Verified	1,434	9,677	14.82

Verification Status	Charge-Off Count	Resolved Count	Charge-Off Rate %
Not Verified	2,141	16,692	12.83

Why This Question Matters

This tests whether income verification actually functions as a risk control — i.e., whether verifying a borrower's income correlates with lower default rates, as one would intuitively expect.

Interpretation
Counterintuitively, "Verified" borrowers show the highest charge-off rate at 16.80%, while "Not Verified" borrowers show the lowest at 12.83% — the opposite of what one might expect if verification were assumed to reduce risk. This suggests verification status is a symptom of risk rather than a safeguard against it: the lender likely verifies income specifically for borrowers whose profiles already raise concern (e.g., high requested amount relative to income, weaker credit signals), while low-risk-looking applicants pass through without scrutiny. This finding implies verification alone is not an effective risk filter, and the lender should examine which underlying borrower characteristics are triggering verification in the first place to better target risk at the source.

Q9 — Borrower Segments Exceeding 20% Charge-Off Rate

```
SELECT grade, chargeoff_count, resolved_count, rate
FROM (
  SELECT
    grade,
    chargeoff_count,
    resolved_count,
    ROUND(chargeoff_count * 100.0::NUMERIC / resolved_count, 2) AS rate
  FROM (
    SELECT
      grade,
      COUNT(CASE WHEN loan_status = 'Charged Off' THEN 1 END) AS chargeoff_count,
      COUNT(CASE WHEN loan_status IN ('Fully Paid', 'Charged Off') THEN 1 END) AS resolved_count
    FROM loans
    WHERE loan_status IN ('Fully Paid', 'Charged Off')
    GROUP BY grade
  ) AS counts
) AS rates
WHERE rate > 20
ORDER BY grade DESC;
```

Result

Grade	Charge-Off Count	Resolved Count	Charge-Off Rate %
G	101	299	33.78
F	319	976	32.68
E	715	2,663	26.85
D	1,118	5,085	21.99

Why This Question Matters

This identifies the exact threshold where borrower segments cross from acceptable to high risk, rather than relying on assumed cutoffs from letter grades alone.

Interpretation
Grades D through G — the bottom half of the rating scale — all exceed the 20% charge-off threshold, with rates climbing from 21.99% to 33.78%, confirming a clear risk cliff between Grade C (17.19%) and Grade D. This signals that Grade D should likely be treated as the practical boundary for high-risk classification in this portfolio, not just E-G as might be assumed from the letter grades alone.

Q10 — Interest Rate: Charged Off vs Fully Paid

```
SELECT
   loan_status,
   ROUND(AVG(int_rate)::NUMERIC, 2) AS avg_int_rate
FROM loans
WHERE loan_status IN ('Fully Paid', 'Charged Off')
GROUP BY loan_status;
```

Result

Loan Status	Avg Interest Rate %
Fully Paid	11.61
Charged Off	13.82

Why This Question Matters

This tests whether interest rate itself correlates with default outcome, independent of any other variable — a basic sanity check on whether risk-based pricing is functioning as designed.

Interpretation
Charged-off loans carry a 2.21 percentage point higher average interest rate (13.82% vs 11.61%) than fully paid loans, confirming that the lender's risk-based pricing system is functioning as intended — riskier borrowers are charged more, and those borrowers do default more often. However, since interest rate is itself derived from grade, this finding largely reinforces what Q6 already showed rather than identifying an independent risk driver.

Q11 — Borrower Income: Charged Off vs Fully Paid

```
SELECT
   loan_status,
   ROUND(AVG(annual_inc)::NUMERIC, 2) AS avg_income
FROM loans
WHERE loan_status IN ('Fully Paid', 'Charged Off')
GROUP BY loan_status;
```

Result

Loan Status	Avg Annual Income
Fully Paid	$69,860.98
Charged Off	$62,421.33

Why This Question Matters

This tests whether borrower income independently predicts default, or whether it's simply correlated with risk factors already captured by grade.

Interpretation

Charged-off borrowers report a lower average annual income ($62,421.33) than fully paid borrowers ($69,860.98), a gap of $7,439.65, or roughly 10.6% in relative terms. While directionally consistent with the expectation that lower income increases default risk, this gap is comparatively modest next to the clear-cut signal seen in Q10's interest rate comparison, and likely reflects the same underlying risk patterns already captured by grade rather than acting as an independent driver of default.

Q12 — DTI: Charged Off vs Fully Paid

```
SELECT
    loan_status,
    ROUND(AVG(dti)::NUMERIC, 2) AS avg_dti
FROM loans
WHERE loan_status IN ('Fully Paid', 'Charged Off')
GROUP BY loan_status;
```

Result

Loan Status	Avg DTI %
Fully Paid	13.15
Charged Off	14.00

Why This Question Matters

This tests whether a borrower's existing debt burden, independent of the new loan, predicts default risk.

Interpretation

Charged-off borrowers show a marginally higher average DTI (14.00%) than fully paid borrowers (13.15%), a gap of only 0.85 percentage points. While directionally consistent with the expectation that higher debt burden increases default risk, this gap is too small to be considered a strong standalone predictor, reinforcing the pattern seen across Q10-Q12: interest rate showed a clear, decisive signal, while income and DTI offer only weak, supporting evidence that likely reflects risk already captured by grade rather than independent drivers.

Q13 — Loan Term Influence on Charge-Off Risk

```
SELECT term, chargeoff_count, resolved_count,
   ROUND(chargeoff_count * 100.0::NUMERIC / resolved_count, 2) AS rate
FROM (
   SELECT
      term,
      COUNT(CASE WHEN loan_status = 'Charged Off' THEN 1 END) AS chargeoff_count,
      COUNT(CASE WHEN loan_status IN ('Charged Off', 'Fully Paid') THEN 1 END) AS resolved_count
   FROM loans
   WHERE loan_status IN ('Charged Off', 'Fully Paid')
   GROUP BY term
) AS charge_off
ORDER BY term;
```

Result

Term	Charge-Off Count	Resolved Count	Charge-Off Rate %
36 months	3,226	29,094	11.09
60 months	2,400	9,481	25.31

Why This Question Matters

This tests whether loan term independently predicts default risk, validating the hypothesis raised in Q4 that longer repayment periods expose borrowers to more life disruptions, increasing default likelihood.

Interpretation

36-month loans have a higher absolute charge-off count (3,226 vs 2,400), but this is misleading on its own since 36-month loans are also a much larger segment overall. The charge-off rate tells the real story — 60-month loans default at 25.31% versus just 11.09% for 36-month loans, a gap of 14.22 percentage points, confirming the prediction made in Q4 that longer-term loans carry disproportionately higher risk due to extended exposure to life disruptions.

Q14 — Higher Interest Rate Loans and Default Rates

```
SELECT rate_band, chargeoff_count, resolved_count,
   ROUND(chargeoff_count * 100.0::NUMERIC / resolved_count, 2) AS rate
FROM (
   SELECT
      CASE
         WHEN int_rate < 10 THEN 'Low (<10%)'
         WHEN int_rate BETWEEN 10 AND 15 THEN 'Medium (10-15%)'
         ELSE 'High (>15%)'
      END AS rate_band,
      COUNT(CASE WHEN loan_status = 'Charged Off' THEN 1 END) AS chargeoff_count,
      COUNT(CASE WHEN loan_status IN ('Charged Off', 'Fully Paid') THEN 1 END) AS resolved_count
   FROM loans
   WHERE loan_status IN ('Charged Off', 'Fully Paid')
   GROUP BY rate_band
) AS bands
ORDER BY rate_band;
```

Result

Rate Band	Charge-Off Count	Resolved Count	Charge-Off Rate %
Low (<10%)	799	12,066	6.62
Medium (10-15%)	2,737	18,513	14.78
High (>15%)	2,090	7,996	26.14

Why This Question Matters

This tests whether interest rate shows a dose-response relationship with default risk — not just a difference in averages (Q10), but a consistent upward trend across rate bands — which would more rigorously confirm rate as a genuine risk signal.

Interpretation

Charge-off rate rises sharply with interest rate band — from 6.62% in the low band (under 10%) to 26.14% in the high band (above 15%), a nearly fourfold increase. This confirms Q10's finding more rigorously: interest rate isn't just correlated with default status on average, it shows a clear dose-response relationship — the higher the rate, the higher the risk, validating that the lender's pricing model meaningfully separates risk tiers.

Q15 — Grades Contributing Most Charge-Offs

```
SELECT grade,
   COUNT(*) AS resolved_count,
   COUNT(CASE WHEN loan_status = 'Charged Off' THEN 1 END) AS chargeoff_count,
   ROUND(COUNT(CASE WHEN loan_status = 'Charged Off' THEN 1 END) * 100.0::NUMERIC /
      (SELECT COUNT(*) FROM loans WHERE loan_status = 'Charged Off'), 2) AS share_of_total_chargeoff
FROM loans
WHERE loan_status IN ('Charged Off', 'Fully Paid')
GROUP BY grade
ORDER BY grade;
```

Result

Grade	Resolved Count	Charge-Off Count	Share of Total Charge-Offs %
A	10,044	602	10.70
B	11,674	1,424	25.31
C	7,834	1,347	23.94
D	5,085	1,118	19.87
E	2,663	715	12.71
F	976	319	5.67
G	299	101	1.80

Why This Question Matters

This distinguishes between charge-off **rate** (how risky a segment is per loan, Q6) and chargeoff **share** (how much a segment actually contributes to total portfolio losses in absolute terms) — a critical distinction for prioritizing risk management resources.

Interpretation
Grade B contributes the largest share of total portfolio charge-offs at 25.31%, despite having a moderate individual charge-off rate of 12.20% (Q6) — driven purely by its large volume (12,019 loans, Q2). In contrast, Grade G has the highest individual charge-off rate at 33.78% but contributes only 1.80% of total losses, since it represents just 0.80% of the portfolio by count. This confirms that rate and absolute contribution are different lenses entirely: a small, high-risk segment can matter less to total dollar losses than a large, moderate-risk one — meaning the lender's real exposure management priority should be Grade B and C, the largest volume segments, not just the highest-rate grades.

Q16 — Purposes Contributing Most Charge-Offs

```
SELECT purpose,
   COUNT(*) AS resolved_count,
   COUNT(CASE WHEN loan_status = 'Charged Off' THEN 1 END) AS chargeoff_count,
   ROUND(COUNT(CASE WHEN loan_status = 'Charged Off' THEN 1 END) * 100.0::NUMERIC /
      (SELECT COUNT(*) FROM loans WHERE loan_status = 'Charged Off'), 2) AS chargeoff_per_purpose
FROM loans
WHERE loan_status IN ('Charged Off', 'Fully Paid')
GROUP BY purpose
ORDER BY chargeoff_per_purpose DESC;
```

Result

Purpose	Resolved Count	Charge-Off Count	Share of Total Charge-Offs %
debt_consolidation	18,055	2,767	49.18
other	3,864	633	11.25
credit_card	5,027	542	9.63
small_business	1,753	474	8.43
home_improvement	2,875	347	6.17
major_purchase	2,150	222	3.95
car	1,499	160	2.84
wedding	926	96	1.71
medical	681	106	1.88
moving	576	92	1.64
house	367	59	1.05
educational	325	56	1.00
vacation	375	53	0.94
renewable_energy	102	19	0.34

Why This Question Matters

This identifies which loan purposes drive the largest absolute losses for the lender, distinguishing volume-driven risk from rate-driven risk — critical for deciding where to focus risk mitigation resources.

Interpretation
Debt consolidation contributes nearly half of all portfolio charge-offs (49.18%), despite carrying only a moderate individual charge-off rate of 15.33% (Q7) — barely above the portfolio average of 14.58%. This dominance is driven entirely by volume: debt consolidation represents 46.94% of all loans in the portfolio (Q3), so even an unremarkable default rate compounds into the single largest contributor to total losses. This mirrors the Grade B pattern seen in Q15 — small_business, despite having the highest individual charge-off rate at 27.04%, contributes only 8.43% to total losses, confirming again that volume, not rate, is the primary driver of absolute portfolio risk exposure.

Q17 — Small Business Risk: Default Rate, Exposure, Contribution

```
SELECT
   ROUND(SUM(funded_amnt)::NUMERIC, 2) AS sb_total_exposure,
   ROUND(COUNT(CASE WHEN loan_status = 'Charged Off' THEN 1 END) * 100.0::NUMERIC /
      (SELECT COUNT(*) FROM loans WHERE loan_status = 'Charged Off'), 2) AS sb_chargeoff_share,
   COUNT(CASE WHEN loan_status = 'Charged Off' THEN 1 END) AS sb_chargeoff_count,
   COUNT(CASE WHEN loan_status IN ('Fully Paid', 'Charged Off') THEN 1 END) AS sb_resolved_count,
   ROUND(COUNT(CASE WHEN loan_status = 'Charged Off' THEN 1 END) * 100.0::NUMERIC /
      COUNT(CASE WHEN loan_status IN ('Fully Paid', 'Charged Off') THEN 1 END), 2) AS
sb_chargeoff_rate
FROM loans
WHERE loan_status IN ('Fully Paid', 'Charged Off') AND purpose = 'small_business';
```

Result

Total Exposure	Share of Total Charge-Offs %	Charge-Off Count	Resolved Count	Charge-Off Rate %
$22,522,675.00	8.43	474	1,753	27.04

Why This Question Matters

This consolidates rate, volume, and dollar exposure into a single view for one specific high-risk segment, testing whether a high individual risk rate translates into meaningful absolute exposure for the lender, or remains contained by low volume.

Interpretation
Small business loans represent $22,522,675 in total funded exposure and carry the highest individual charge-off rate of any purpose category at 27.04% — over 1.8x the portfolio average of 14.58%. Despite this elevated risk, small business loans contribute only 8.43% to total portfolio charge-offs, since their relatively low volume (1,753 resolved loans) limits their absolute impact on the book. This combination suggests the lender should not necessarily exit small business lending, but should consider tightening underwriting criteria or repricing this specific category, since its risk is real and measurable but currently contained by limited volume — a containment that would disappear if small business lending were scaled up without addressing the underlying default rate.

Q18 — Grade-Purpose Combinations with Highest Risk Exposure

```
SELECT grade, purpose,
   COUNT(CASE WHEN loan_status = 'Charged Off' THEN 1 END) AS chargeoff_count,
   COUNT(CASE WHEN loan_status IN ('Charged Off', 'Fully Paid') THEN 1 END) AS resolved_count,
   ROUND(COUNT(CASE WHEN loan_status = 'Charged Off' THEN 1 END) * 100.0::NUMERIC /
      COUNT(CASE WHEN loan_status IN ('Charged Off', 'Fully Paid') THEN 1 END), 2) AS chargeoff_rate
FROM loans
WHERE loan_status IN ('Charged Off', 'Fully Paid')
GROUP BY grade, purpose
HAVING COUNT(CASE WHEN loan_status IN ('Charged Off', 'Fully Paid') THEN 1 END) >= 100
ORDER BY chargeoff_rate DESC;
```

Result (Top 10)

Grade	Purpose	Charge-Off Count	Resolved Count	Charge-Off Rate %
F	debt_consolidation	199	556	35.79
E	small_business	66	188	35.11
G	debt_consolidation	55	162	33.95
D	small_business	94	285	32.98
E	other	66	210	31.43
C	small_business	97	340	28.53
E	debt_consolidation	406	1,494	27.18
B	small_business	120	456	26.32
D	other	124	479	25.89
E	credit_card	63	262	24.05

Why This Question Matters

Single-dimension analysis (grade alone, purpose alone) can mask compounding risk effects. This tests whether specific grade-purpose combinations carry risk beyond what either factor predicts independently — critical for identifying the lender's true highest-risk segments. A minimum threshold of 100 resolved loans was applied to exclude statistically unreliable small samples (combinations with under 10 loans showed wild swings between 0% and 100%).

Interpretation

When grade and purpose are combined, the highest-risk segments emerge as Grade F debt consolidation loans (35.79%), Grade E small business loans (35.11%), and Grade G debt consolidation loans (33.95%) — filtering to combinations with at least 100 resolved loans to exclude statistically unreliable small samples. Critically, these combined rates exceed what either dimension predicts alone: Grade F's standalone charge-off rate is 32.68% (Q6) and debt consolidation's standalone rate is 15.33% (Q7), yet together they compound to 35.79%, higher than either factor in isolation. This demonstrates that risk is not simply additive across borrower characteristics — certain grade-purpose combinations carry compounding risk that single-dimension analysis would underestimate, meaning the lender's risk models should account for interaction effects between grade and purpose, not just each factor independently.

Q19 — Rank Borrower Segments by Portfolio Risk Exposure

```
SELECT grade,
   COUNT(CASE WHEN loan_status = 'Charged Off' THEN 1 END) AS chargeoff_count,
   COUNT(CASE WHEN loan_status IN ('Charged Off', 'Fully Paid') THEN 1 END) AS resolved_count,
   ROUND(COUNT(CASE WHEN loan_status = 'Charged Off' THEN 1 END) * 100.0::NUMERIC /
      (SELECT COUNT(*) FROM loans WHERE loan_status = 'Charged Off'), 2) AS
portfolio_chargeoff_share,
   DENSE_RANK() OVER (ORDER BY
      ROUND(COUNT(CASE WHEN loan_status = 'Charged Off' THEN 1 END) * 100.0::NUMERIC /
      (SELECT COUNT(*) FROM loans WHERE loan_status = 'Charged Off'), 2) DESC
   ) AS risk_rank
FROM loans
WHERE loan_status IN ('Charged Off', 'Fully Paid')
GROUP BY grade;
```

Result

Grade	Charge-Off Count	Resolved Count	Portfolio Share %	Risk Rank
B	1,424	11,674	25.31	1
C	1,347	7,834	23.94	2
D	1,118	5,085	19.87	3
E	715	2,663	12.71	4
A	602	10,044	10.70	5
F	319	976	5.67	6
G	101	299	1.80	7

Why This Question Matters

This consolidates every prior charge-off finding (rate, count, share) into a single definitive ranking of which grades the lender should prioritize when managing total portfolio risk — answering the question Section 4 was built to resolve.

Interpretation

The final ranking confirms Grade B as the single largest contributor to total portfolio losses (25.31%, rank 1), followed by Grade C (23.94%) and Grade D (19.87%) — not because these grades carry the highest individual risk, but because they represent the largest volume segments in the portfolio. Grade G, despite having the highest individual charge-off rate of any segment (33.78%, Q6), ranks last in total contribution (1.80%) simply because its small volume (299 loans) limits its absolute impact. This consolidated ranking confirms the central lesson of Section 4: a lender managing portfolio risk should prioritize attention on high-volume, moderate-risk segments like Grade B and C, since they drive the majority of dollar losses — not just chase the highest individual rates in small, already-contained segments like Grade G.

Q20 — Rank Grades by Risk Exposure: ROW_NUMBER vs RANK vs DENSE_RANK

```
SELECT grade,
   COUNT(CASE WHEN loan_status = 'Charged Off' THEN 1 END) AS chargeoff_count,
   COUNT(CASE WHEN loan_status IN ('Charged Off', 'Fully Paid') THEN 1 END) AS resolved_count,
   DENSE_RANK() OVER (ORDER BY
      ROUND(COUNT(CASE WHEN loan_status = 'Charged Off' THEN 1 END) * 100.0::NUMERIC /
      (SELECT COUNT(*) FROM loans WHERE loan_status = 'Charged Off'), 2) DESC
   ) AS risk_dense_rank,
   RANK() OVER (ORDER BY
      ROUND(COUNT(CASE WHEN loan_status = 'Charged Off' THEN 1 END) * 100.0::NUMERIC /
      (SELECT COUNT(*) FROM loans WHERE loan_status = 'Charged Off'), 2) DESC
   ) AS risk_rank,
   ROW_NUMBER() OVER (ORDER BY
      ROUND(COUNT(CASE WHEN loan_status = 'Charged Off' THEN 1 END) * 100.0::NUMERIC /
      (SELECT COUNT(*) FROM loans WHERE loan_status = 'Charged Off'), 2) DESC
   ) AS risk_row_number
FROM loans
WHERE loan_status IN ('Fully Paid', 'Charged Off')
GROUP BY grade;
```

Result

Grade	Charge-Off Count	Resolved Count	Dense Rank	Rank	Row Number
B	1,424	11,674	1	1	1
C	1,347	7,834	2	2	2
D	1,118	5,085	3	3	3
E	715	2,663	4	4	4
A	602	10,044	5	5	5
F	319	976	6	6	6
G	101	299	7	7	7

Why This Question Matters

This tests technical understanding of the three ranking functions — specifically, recognizing when their behaviour converges (no ties) versus diverges (ties present), a common interview question.

Interpretation

Running ROW_NUMBER(), RANK(), and DENSE_RANK() side by side on grade-level charge-off share produces identical results across all seven grades, since none of the portfolio_chargeoff_share values are tied — each grade has a distinct percentage. This confirms that the three functions only diverge in behaviour when tied values are present in the data: ROW_NUMBER() would still assign unique sequential numbers to tied rows, RANK() would assign the same rank to ties but skip subsequent numbers, and DENSE_RANK() would assign the same rank to ties without skipping. For this particular dataset, DENSE_RANK() (used in Q19) remains the safest default choice for production use, since it would handle ties gracefully if they existed, even though this specific grade-level breakdown has none.

Q21 — Compare Segment Default Rate to Portfolio Average (Window Functions)

```
SELECT
    grade,
    chargeoff_count,
    resolved_count,
    ROUND(chargeoff_count * 100.0::NUMERIC / resolved_count, 2) AS grade_rate,
    ROUND(SUM(chargeoff_count) OVER () * 100.0::NUMERIC / SUM(resolved_count) OVER (), 2) AS
portfolio_avg_rate,
    ROUND((chargeoff_count * 100.0::NUMERIC / resolved_count) -
        (SUM(chargeoff_count) OVER () * 100.0::NUMERIC / SUM(resolved_count) OVER ()), 2) AS
rate_vs_avg
FROM (
    SELECT
        grade,
        COUNT(CASE WHEN loan_status = 'Charged Off' THEN 1 END) AS chargeoff_count,
        COUNT(CASE WHEN loan_status IN ('Charged Off', 'Fully Paid') THEN 1 END) AS resolved_count
    FROM loans
    WHERE loan_status IN ('Charged Off', 'Fully Paid')
    GROUP BY grade
) AS counts
ORDER BY grade;
```

Result

Grade	Charge-Off Count	Resolved Count	Grade Rate %	Portfolio Avg %	Rate vs Avg
A	602	10,044	5.99	14.58	-8.59
B	1,424	11,674	12.20	14.58	-2.39
C	1,347	7,834	17.19	14.58	+2.61
D	1,118	5,085	21.99	14.58	+7.40
E	715	2,663	26.85	14.58	+12.26
F	319	976	32.68	14.58	+18.10
G	101	299	33.78	14.58	+19.19

Why This Question Matters

This pinpoints exactly where each grade sits relative to the portfolio's own benchmark, using window functions to broadcast a portfolio-wide average alongside each grade's individual rate — without collapsing the result into a single summary row.

Interpretation

Only Grades A (5.99%) and B (12.20%) perform better than the portfolio average charge-off rate of 14.58%, while every other grade — C through G — exceeds it, with deviations ranging from +2.61 percentage points (Grade C) to +19.19 percentage points (Grade G). The portfolio average itself falls between Grade B (12.20%) and Grade C (17.19%), establishing a sharper dividing line than the 20% threshold identified in Q9: the true split between "above-average" and "below-average" risk performance sits right at the B/C boundary, meaning the lender's effective safe zone is narrower than the letter grades alone might suggest — only the top two grades genuinely outperform the book's own average.

Q22 — Identify Above-Average Risk Segments (Subquery + Aggregation)

```
SELECT *
FROM (
   SELECT
      grade,
      chargeoff_count,
      resolved_count,
      ROUND(chargeoff_count * 100.0::NUMERIC / resolved_count, 2) AS grade_rate,
      ROUND(SUM(chargeoff_count) OVER () * 100.0::NUMERIC / SUM(resolved_count) OVER (), 2) AS
portfolio_avg_rate,
      ROUND((chargeoff_count * 100.0::NUMERIC / resolved_count) -
         (SUM(chargeoff_count) OVER () * 100.0::NUMERIC / SUM(resolved_count) OVER ()), 2) AS
rate_vs_avg
   FROM (
      SELECT
         grade,
         COUNT(CASE WHEN loan_status = 'Charged Off' THEN 1 END) AS chargeoff_count,
         COUNT(CASE WHEN loan_status IN ('Charged Off', 'Fully Paid') THEN 1 END) AS resolved_count
      FROM loans
      WHERE loan_status IN ('Charged Off', 'Fully Paid')
      GROUP BY grade
   ) AS counts
) AS q21_data
WHERE rate_vs_avg > 0
ORDER BY rate_vs_avg DESC;
```

Result

Grade	Charge-Off Count	Resolved Count	Grade Rate %	Portfolio Avg %	Rate vs Avg
G	101	299	33.78	14.58	+19.19
F	319	976	32.68	14.58	+18.10
E	715	2,663	26.85	14.58	+12.26
D	1,118	5,085	21.99	14.58	+7.40
C	1,347	7,834	17.19	14.58	+2.61

Why This Question Matters

This produces a focused, action-ready view — isolating only the segments that underperform the portfolio average, ranked by severity, rather than requiring a manager to scan the full grade breakdown to identify problem areas.

Interpretation

Five grades — C, D, E, F, and G — qualify as above-average risk segments, with Grade G showing the largest deviation at +19.19 percentage points above the portfolio average of 14.58%, followed closely by Grade F (+18.10) and Grade E (+12.26). While this is the same underlying data as Q21, filtering and sorting specifically for above-average segments gives a portfolio manager a focused action list — rather than scanning all seven grades, they can immediately see which five segments require intervention, ranked by severity, making this the practical reporting view a risk team would actually use day to day.

Q23 — Portfolio Risk Scorecard

```
SELECT grade,
   chargeoff_count,
   resolved_count,
   ROUND(chargeoff_count * 100.0::NUMERIC / resolved_count, 2) AS portfolio_chargeoff_rate,
   ROUND(SUM(chargeoff_count) OVER () * 100.0::NUMERIC / SUM(resolved_count) OVER (), 2) AS
portfolio_avg_chargeoff_rate,
   ROUND((chargeoff_count * 100.0::NUMERIC / resolved_count)
      - (SUM(chargeoff_count) OVER () * 100.0::NUMERIC / SUM(resolved_count) OVER ()), 2) AS
avg_vs_rate,
   CASE
      WHEN ROUND(chargeoff_count * 100.0::NUMERIC / resolved_count, 2) < 14.58 THEN 'Low Risk'
      WHEN ROUND(chargeoff_count * 100.0::NUMERIC / resolved_count, 2) < 20 THEN 'Moderate Risk'
      ELSE 'High Risk'
   END AS risk_band
FROM (
   SELECT grade,
      COUNT(CASE WHEN loan_status = 'Charged Off' THEN 1 END) AS chargeoff_count,
      COUNT(CASE WHEN loan_status IN ('Charged Off', 'Fully Paid') THEN 1 END) AS resolved_count
   FROM loans
   WHERE loan_status IN ('Charged Off', 'Fully Paid')
   GROUP BY grade
) AS counts
ORDER BY avg_vs_rate;
```

Result

Grade	Charge-Off Count	Resolved Count	Rate %	Portfolio Avg %	vs Avg	Risk Band
A	602	10,044	5.99	14.58	-8.59	Low Risk
B	1,424	11,674	12.20	14.58	-2.39	Low Risk
C	1,347	7,834	17.19	14.58	+2.61	Moderate Risk
D	1,118	5,085	21.99	14.58	+7.40	High Risk
E	715	2,663	26.85	14.58	+12.26	High Risk
F	319	976	32.68	14.58	+18.10	High Risk
G	101	299	33.78	14.58	+19.19	High Risk

Why This Question Matters

This consolidates every threshold established across the project (Q9's 20% cliff, Q21's average comparison) into a single, decision-ready scorecard — the kind of summary table an actual risk team would use to monitor the portfolio.

Interpretation

The risk scorecard confirms a clear three-tier structure: Low Risk (Grades A, B) perform below the portfolio average of 14.58%, by margins of 8.59 and 2.39 percentage points respectively. Moderate Risk (Grade C) sits just above average at 17.19%. High Risk (Grades D through G) all exceed 20%, climbing as high as 33.78% for Grade G. The lender should treat Grade D as the practical underwriting boundary, tighten approval criteria for small business loans specifically (Q17-Q18), and continue monitoring Grade B and C despite their "lower risk" classification here, since they remain the largest absolute contributors to total portfolio losses (Q15, Q19).